Help me understand

Enlargements

Exam code: 0607

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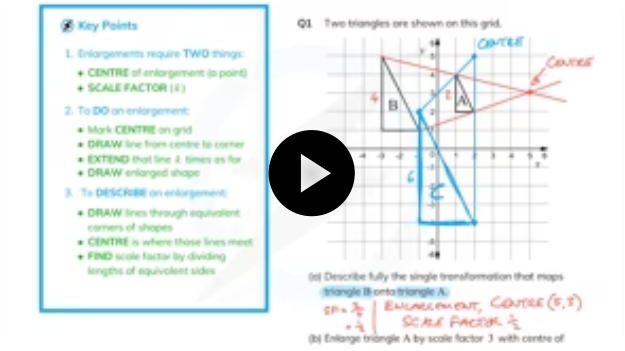

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Enlargements

What is an enlargement?

- An **enlargement changes** the **size** and **position** of a shape
- The **length of each side** of the shape is **multiplied** by a **scale factor**
  - If the scale factor is **greater than 1** then the **enlarged image** will be **bigger** than the **original object**
  - If the scale factor is **between 0 and 1 (fractional)** then the **enlarged image** will be **smaller** than the **original object**

- The **centre of enlargement** determines the **position** of the enlarged image
  - If the scale factor is **greater than 1** then the **enlarged image** will be **further away** from the centre of enlargement
  - If the scale factor is **between 0 and 1** then the **enlarged image** will be **closer to** the centre of enlargement

How do I enlarge a shape?

- STEP 1
  Pick a **vertex** of the shape and count the **horizontal and vertical distances** from the **centre of enlargement**
- STEP 2
  Multiply **both** the horizontal and vertical distances by the

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

distances to find the **enlarged vertex**

- **STEP 4**
  Repeat the steps for the **other vertices**
  - You might be able to draw the enlarged shape from the first vertex by **multiplying the original lengths** by the scale factor
    - This can be done **quickly** if the shape is made up of **vertical and horizontal lines**

- **STEP 5**
  Connect the vertices on the **enlarged image** and label it

Examiner Tips and Tricks

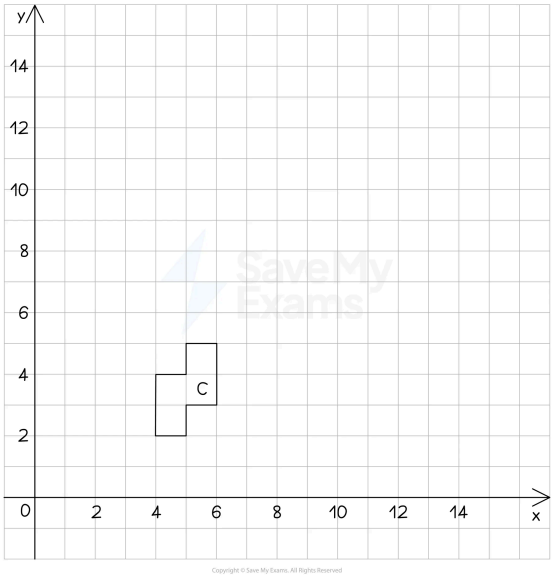
To **check** that you have enlarged a shape **correctly**:

- **Draw lines** going **from the centre of enlargement** to **each of the vertices** of the original shape
- **Extend** these lines
- The **lines should go through** the **corresponding vertices** of the enlarged image

Worked Example

(a) On the grid below enlarge shape C using scale factor 2 and centre of enlargement (2, 1).

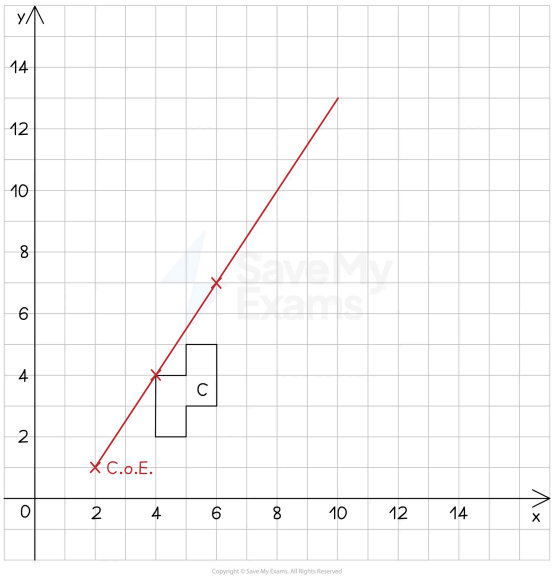
Label your enlarged shape C'.



Answer:

Start by marking on the centre of enlargement (CoE).
Count the number of squares in both a horizontal and vertical

become 4 to the right and 6 up.
Count these new distances from the CoE to the corresponding point on the enlarged image and mark it on.
Draw a line through the CoE and the pair of corresponding points, they should line up in a straight line.



Repeat this process for each of the vertices on the original object (or at least 2).
Join adjacent vertices on the enlarged image as you go.
Label the enlarged image C'.

(b) On the grid below enlarge shape C using scale factor $\frac{1}{2}$ and centre of enlargement (4, 2).

Write down the four vertices of your enlarged shape.

Answer:

Mark the centre of enlargement at (4, 2)
Count the number of squares horizontally and vertically to any vertex - we've chosen the vertex at (-2, 4)

Multiply these distance by the scale factor, $\frac{1}{2}$

$$6 \text{ 'right'} \times \frac{1}{2} = 3 \text{ 'right'}$$

$$2 \text{ 'down'} \times \frac{1}{2} = 1 \text{ 'down'}$$

Count these new distances (which should be smaller than the originals) from the CoE to find the corresponding point on the new image and mark it on

Repeat as required and draw lines through corresponding vertices and the CoE as a check

The four vertices of the enlarged shape are (0, 0), (0, 4), (1, 3) and (1, 1)

How do I describe an enlargement?

- To describe an **enlargement**, you must:
  - State that the transformation is an **enlargement**
  - State the **scale factor**
    - This may be an integer or a fraction
  - Give the coordinates of the **centre of enlargement**
- To find the **scale factor**:
  - **Pick a side** of the **original shape**
  - Identify the **corresponding side** on the **enlarged image**
    - For a fractional enlargement, the side on the enlarged image will be **smaller** than the corresponding side on the original image
  - **Divide** the **length of the enlarged side** by the **length of the original side**
- To find the **centre of enlargement**:
  - **Pick a vertex** of the **original shape**
  - Identify the **corresponding vertex** on the **enlarged image**
  - Draw a **line going through these two vertices**
  - **Repeat** this for the **other vertices** of the original shape
  - These lines will **intersect** at the centre of **enlargement**

How do I reverse an enlargement?

- If a shape has been **enlarged**, you can perform a single transformation to **return the shape** to its **original size and position**
- An enlargement can be **reversed** by multiplying the enlarged shape by the **reciprocal of the original scale factor**
  - The centre of enlargement is the same

- an enlargement of scale factor $\frac{1}{3}$
- with centre of enlargement (-1, 6)

Worked Example

Describe fully the single transformation that creates shape B from shape A.

Answer:

We can see that the image is larger than the original object, therefore it must be an **enlargement.**
As the enlarged image is bigger than the original object, the scale factor must be greater than 1.
Compare two corresponding edges on the object and the image to find the scale factor.

The height of the original "H" is 3 squares
The height of the enlarged "H" is 9 squares

$$\therefore \text{Scale Factor} = \frac{9}{3} = 3$$

Draw a straight line through the CoE and a pair of corresponding points on the original object and the enlarged image.
Repeat this step for as many vertices as you feel you need to so you can confidently locate the CoE.

Shape A has been enlarged using a scale factor of 3 and a centre of enlargement (9, 7) to create shape B

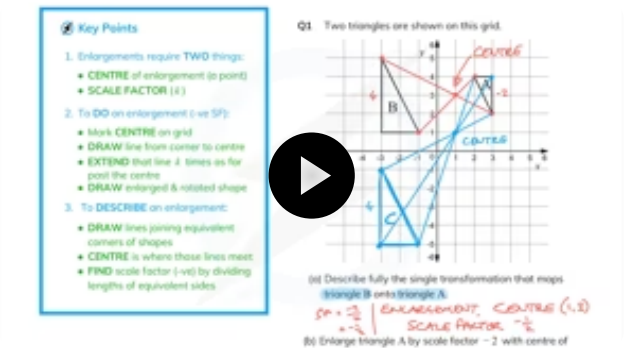
(b) Describe fully the single transformation that creates shape B from shape A.

Answer:

We can see the image is smaller than the original so it is a **fractional enlargement**
Compare two corresponding edges to find the scale factor - we've used the top edge

Draw straight lines through corresponding vertices on the original shape
Repeat this 3–4 times and you should find the lines intersect at the same point
This point will be the CoE

Shape A has been enlarged using a scale factor of $\frac{1}{3}$ and a centre of enlargement (3, −3.5) to create shape B



Did this video help you? Yes No

Negative enlargements

How do I enlarge a shape if it has a negative scale factor?

- You will still need to **perform enlargements** with **negative scale factors**
  - it is possible but unusual to be asked to **identify** one

by 180°
    - the **distance** between the **centre of enlargement (CoE) and the enlarged image,** is measured on the **opposite side of the CoE**

How do I reverse a negative enlargement?

- If a shape has undergone a **negative enlargement**, you can perform a single transformation to **return the shape** to its **original size and position**
- An enlargement can be **reversed** by multiplying the enlarged shape by the **reciprocal of the original scale factor**
    - the sign of the scale factor remains negative
    - the centre of enlargement is the same

- For a shape enlarged by a scale factor of $-\frac{1}{2}$ with centre of enlargement (0, 5)
    - The **reverse transformation** is
        - an enlargement of scale factor -2
        - with centre of enlargement (0, 5)

Examiner Tips and Tricks

Draw lines **through the CoE** and a **vertex on the original object.**

This will remind you that the distances away from the CoE carry on in the **opposite direction** for a negative scale factor.

Examiner Tips and Tricks

Watch out, exam questions are quite keen on combining both **negative** and **fractional** scale factors!

Worked Example

On the grid below enlarge shape F using scale factor $-\frac{1}{3}$ and centre of enlargement $(6, -1)$.

Label this shape F'.

If the area of F is 45 cm^2 write down the area of F'.

Answer:

Start by marking the centre of enlargement (CoE) (6, -1) and selecting a starting vertex

Count the horizontal and vertical distances from the vertex to the CoE
Multiply those distances by the scale factor

Vertex at (-4, 3)

Distance to CoE from vertex on original object: 3 to the right and 3 up

Distances from CoE to corresponding vertex on enlarged image: $3 \times \frac{1}{3} = 1$ to the right and $3 \times \frac{1}{3} = 1$ up

Count the new distances from the CoE, on the other side from the original object
Mark on the position of the corresponding point on the enlarged image

Draw a straight line through the corresponding vertices and the CoE to check that they line up

Repeat this process for each vertex in turn

Connect the vertices as you go around so that you don't forget which should connect to which

Remember, your enlarged image will be rotated by 180°

The length scale factor is $\frac{1}{3}$, meaning that each edge of the enlarged image is $\frac{1}{3}$ the length of the corresponding edge on the original object

Find the area scale factor by squaring the length scale factor

$$\text{Area scale factor} = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

$45 \times \frac{1}{9}$

$5\ cm^2$

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